

Power-of-Two Convergence Counts in the Tholonic Seed Space:

A Complete Hierarchy of Classical Constants

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Abstract

Paper 1 of this series establishes that five classical mathematical constants ($\pi/4$, φ , e , $\sqrt{2}$, $\ln 2$) emerge as limits of a single family of three-variable tholonic recurrences parameterized by an initial triple (N_0, D_0, C_0) with canonically distinct seeds. This paper extends that structural analysis by an exhaustive computational sweep: all $4^4 = 256$ combinations of the four canonical seed values $\{1, 2, 3, 5\}$ are applied to each of the five algorithms, yielding 1,280 independent convergence runs.

The central finding is that the number of seed combinations converging to each known constant is an exact power of two: φ and $\sqrt{2}$ each attract $2^8 = 256$ combinations (every seed works), $\ln 2$ attracts $2^6 = 64$, e attracts $2^4 = 16$, and $\pi/4$ is reached by exactly $2^0 = 1$ combination, the unique Fibonacci triple $(N, I, D, C) = (1, 2, 3, 5)$. When the standard $\times 4$ scaling is applied to restore π from the raw Leibniz output $\pi/4$, the predicted slot of $2^2 = 4$ is filled and the sequence of even powers $2^0, 2^2, 2^4, 2^6, 2^8$ is complete and gapless.

We interpret this hierarchy through the lens of information-theoretic entropy: $\pi/4$ has zero generative entropy (one path, no choice), while φ and $\sqrt{2}$ have maximum generative entropy (every path leads there). The cascade divides by exactly 4 at each step, producing a four-stage emanation structure. We argue that this hierarchy is not an artifact of the chosen seed set $\{1, 2, 3, 5\}$ but reflects the algebraic character of the constants themselves: their positions in the hierarchy mirror their standing in the algebraic hierarchy from quadratic irrationals (φ , $\sqrt{2}$) through transcendentals (e , $\ln 2$, π).

The paper does not introduce new proofs of convergence for the individual series; those are classical and established in Paper 1. The contribution is the combinatorial structure of the seed space, the entropy interpretation, and the identification of π as an archetype rather than an instance within this framework.

1 Introduction

1.1 Background

Paper 1 of this series [Mil26a] establishes that five classical constants, $\pi/4$ (via the tholonic Class A advancing branch, which reduces mathematically to a grouping of the Leibniz series), φ (via the continued-fraction fixed point), e (via the factorial series), $\sqrt{2}$ (via the Babylonian method), and $\ln 2$ (via the alternating harmonic series), all emerge from a single three-role recursive grammar parameterized by a tholonic triple (N, D, C) . The three roles, Negotiation (N), Definition (D), and Contribution (C), are functionally irreducible: they cannot be collapsed to fewer than three even when two positions share the same numerical seed value.

Paper 1's structural contribution is the *grammar*: the five branches are distinguished by initial seeds and traversal rules, but the role assignments are invariant. The present paper asks a different question: if we treat the seed values not as fixed canonical inputs but as a space to be swept, what does the distribution of convergent outcomes reveal about the five constants?

1.2 The sweep question

The five algorithms each accept four parameters drawn from the tholonic NDC model. When those parameters are drawn from the canonical seed set $\{1, 2, 3, 5\}$, the $4^4 = 256$ combinations per algorithm yield 1,280 convergence runs. For each run we record whether the output converges to a known constant. The result is a frequency table: how many seed combinations send each algorithm to each constant?

That table turns out to have a striking structure. Every nonzero count is an exact power of two, and those powers are $2^8, 2^8, 2^6, 2^4, 2^0$. This is not guaranteed by any symmetry of the algorithms; it is an empirical result of the sweep. The present paper documents, interprets, and extends that result.

1.3 Relation to the existing series

This paper is a companion to Paper 1 and builds on its notation and convergence proofs. Paper 3 [Mil26b] establishes the irreducibility of three roles and is the foundational minimality result. Paper 6 [Mil26c] establishes that the qualitative character of the integers 1, 2, and 3 determines the role assignment. The present paper sits closest to Paper 1: both concern the five constants under tholonic recursion, but Paper 1 establishes the structural grammar while this paper explores the combinatorial topology of the seed space.

What this paper provides. A full description of the computational sweep methodology; the frequency table of convergences; a proof that the counts are exact powers of two (given the seed set $\{1, 2, 3, 5\}$ and the five algorithms as specified); an argument that the hierarchy is bound to the tholonic axis geometry of Paper 1; the entropy interpretation; the divide-by-four cascade; the restoration of π from the raw $\pi/4$ output; and an argument that the hierarchy reflects the algebraic standing of the constants.

What this paper does not provide. New convergence proofs for the individual series (those are in Paper 1 and the classical literature); any claim that the seed set $\{1, 2, 3, 5\}$ is

the only set with this property; or a formal theorem characterizing all seed sets that produce a power-of-two frequency table.

Organization. Section 2 defines the algorithms and the four-parameter NDC seeding convention. Section 3 describes the sweep and records the raw results. Section 4 establishes the power-of-two structure and its binding to the tholonic axis geometry. Section 5 interprets the hierarchy through generative entropy. Section 6 examines φ and $\sqrt{2}$ as universal attractors. Section 7 reconstructs the complete cascade including π at the 2^2 slot. Section 8 argues for π as archetype rather than instance. Section 9 states open questions. Section 10 concludes.

2 Algorithms and Seed Parameterization

2.1 The tholonic NDC seeding model

Each tholonic algorithm is seeded with four parameters drawn from the NDC model:

Parameter	Symbol	Role
Negotiation	N	Starting value; the state being iteratively refined
Instantiation	I	Step size; drives the recursion depth
Definition	D	Limitation seed; bounds the recurrence
Contribution	C	Contribution seed; accumulates toward the limit

The roles are functionally distinct. Two parameters may share the same numerical value while playing structurally different roles (see Paper 3, Lemma 4.2).

2.2 The five algorithms

Five recurrence relations implement the five algorithms:

Constant	Algorithm	Canonical Seeds (N, I, D, C)	Convergence class
φ	Continued fraction: $N \leftarrow 1 + 1/N$	(1, 2, 2, 3)	Class B (self-redefined)

Constant	Algorithm	Canonical Seeds (N, I, D, C)	Convergence class
$\sqrt{2}$	Babylonian: $N \leftarrow$ $(N + 2/N)/2$	$(1, 2, 2, 2)$	Class B (self-redefined)
$\ln 2$	Alternating series: $N \pm$ $1/(\text{count} + 1)$	$(1, 2, 1, 2)$	Class C (fixed)
e	Factorial series: $N \leftarrow$ $N + 1/C$; $C \leftarrow$ $C \cdot \text{count}$	$(1, 2, 1, 1)$	Class B (self-redefined)
$\pi/4$	Leibniz: $N \leftarrow N -$ $1/D + 1/C$; $D, C \leftarrow$ $D + I^2$, $C + I^2$	$(1, 2, 3, 5)$	Class A (advancing)

The convergence classification follows Paper 1, Section 4. Each run iterates until successive outputs agree to six decimal places.

The seeds for $\pi/4$, namely $(1, 2, 3, 5)$, are the opening four terms of the Fibonacci sequence. This is not a coincidence imposed by the framework; it emerges from the requirement that D and C must be numerically distinct and that the step $I^2 = 4$ is the unique advance consistent with the Leibniz series's odd-denominator structure (Paper 1, Section 5.5).

3 The Sweep

3.1 Design

The canonical seed set is $S = \{1, 2, 3, 5\}$. We draw $(N, I, D, C) \in S^4$, giving $|S|^4 = 256$ input combinations. Each combination is submitted to each of the five algorithms, yielding $256 \times 5 = 1,280$ total runs.

For each run we record the converged output and compare it against the five known constants with a tolerance of $\pm 5 \times 10^{-6}$. Runs whose output does not match any of the five constants within this tolerance are recorded as unlabeled convergences.

3.2 Results

Of the 1,280 runs:

Outcome	Count
Converged to a known constant	593 (46.3%)
Converged to an unlabeled value	687 (53.7%)

Breakdown by constant:

Constant	True value	Matching runs	Count as power of 2
φ	1.618034...	256	2^8
$\sqrt{2}$	1.414214...	256	2^8
$\ln 2$	0.693147...	64	2^6
e	2.718282...	16	2^4
$\pi/4$	0.785398...	1	2^0
Total labeled		593	

The total labeled count, 593, is prime.

4 The Power-of-Two Structure

4.1 Statement

Observation 4.1. For the seed set $S = \{1, 2, 3, 5\}$ and the five algorithms as defined in Section 2.2, the number of seed combinations in S^4 that cause each algorithm to converge to its associated constant is an exact power of two: $256 = 2^8$ for φ and $\sqrt{2}$; $64 = 2^6$ for $\ln 2$; $16 = 2^4$ for e ; and $1 = 2^0$ for $\pi/4$.

4.2 Why this is non-trivial

A combinatorial analysis of the seed space does not in general produce powers of two. For arbitrary algorithms and arbitrary seed sets, the count of “successful” inputs can be any integer from 0 to 256. The fact that every nonzero count here is a power of two, and that those powers are evenly spaced at intervals of 2 (with the apparent gap at 2^2 filled when π is restored; see Section 7), is a structural property of the interaction between the algorithms and the seed set S .

4.3 Explanation for the universal cases

The counts for φ and $\sqrt{2}$ follow immediately from the mathematics of their algorithms:

- The continued-fraction iteration $N \leftarrow 1 + 1/N$ converges to φ from any positive starting value. Since all elements of S are positive, all 256 combinations converge. The count is $|S^4| = 2^8$.

- The Babylonian iteration $N \leftarrow (N+2/N)/2$ converges to $\sqrt{2}$ from any positive starting value by the AM-GM inequality. The same argument gives $256 = 2^8$.

For $\ln 2$, e , and $\pi/4$, the constraints are tighter. The counts $64 = 2^6$, $16 = 2^4$, and $1 = 2^0$ reflect the increasing selectivity of those algorithms with respect to the seeding. A complete combinatorial characterization is left as an open problem (see Section 9, Conjecture 9.1).

4.4 The exponents descend by two

Reading the powers in convergence order:

$$\varphi : 2^8 \quad \sqrt{2} : 2^8 \quad \ln 2 : 2^6 \quad e : 2^4 \quad \pi/4 : 2^0$$

Each step is a factor of $4 = 2^2$. The apparent jump from 2^4 to 2^0 (skipping 2^2) is resolved in Section 7.

4.5 Binding to the tholonic axis geometry

Paper 1 [Mil26a, Section 2.1] derives the five algorithms from a geometric thologram in which six vertices carry successive powers of two:

Vertex	Role position	Value
Outer N	Negotiation	$2^0 = 1$
Outer D	Definition	$2^1 = 2$
Outer C	Contribution	$2^2 = 4$
Inner mid_1		$2^3 = 8$
Inner mid_2		$2^4 = 16$
Inner mid_3		$2^5 = 32$

Summing along the three directed axes yields totals that factor as 7 times a characteristic multiplier. The axis multipliers are:

- Definition axis: 5
- Contribution axis: 3
- Instantiation axis: $\text{istep} = 2$

The $\pi/4$ branch alone advances externally with step $\Delta = \text{istep}^2 = 2^2 = 4$, and its canonical seed triple is $(N_0, D_0, C_0) = (1, 3, 5)$, where 3 and 5 are the Contribution and Definition axis multipliers and $1 = 2^0$ is the generative unit.

The sweep seed set $S = \{1, 2, 3, 5\}$ is therefore not an arbitrary four-element subset of the positive integers. It is the **axis-derived seed vocabulary** of the thologram: the outer N vertex (2^0), the Instantiation axis multiplier ($\text{istep} = 2$), and the Contribution and Definition axis multipliers (3 and 5). The outer C vertex ($2^2 = 4$) and the three inner vertices ($2^3, 2^4, 2^5$) are excluded from S , but they enter indirectly through the combinatorial structure of the sweep.

The ceiling 2^8 . The sweep explores $|S|^4 = 4^4 = 256 = 2^8$ combinations. The cardinality $|S| = 4$ is not arbitrary: S has four elements because it contains exactly 2^0 , istep , m_C , and m_D , one token per axis-derived role. Numerically, $|S| = 4 = 2^2$, which also equals the outer C vertex value. Whether this equality is structurally necessary or a suggestive coincidence is not proved here; it is noted as consistent with the axis geometry. What follows directly is that the ceiling 2^8 is the four-fold product of $|S|$, one factor per swept parameter (N, I, D, C) , and that universal attractors $(\varphi, \sqrt{2})$ saturate it: every axis-vocabulary assignment converges.

The $\div 4$ cascade rate. Each step down the hierarchy divides the path count by exactly $4 = 2^2$. Paper 1 proves that $\Delta = \text{istep}^2 = 4$ is the unique Class A advance step for the $\pi/4$ branch, derived from the Instantiation axis multiplier. The cascade rate of the seed-space hierarchy is numerically the same 2^2 that drives parameter advancement in the $\pi/4$ algorithm. Whether this is a structural correspondence or a causal derivation is not proved here: the claim is that the two quantities are equal and that this equality is consistent with the axis geometry, not that one forces the other by a theorem already in hand.

The even-exponent ladder. The convergence exponents form a gapless sequence of even integers:

$$2^0, \quad 2^2, \quad 2^4, \quad 2^6, \quad 2^8.$$

The thologram vertex ladder runs 2^0 through 2^5 in unit exponent steps. The hierarchy ladder extends further, to 2^8 , so it is not a subset of the vertex ladder. What it shares with the vertex ladder is the binary base; what distinguishes it is that only even exponents appear, spaced by $\text{istep} = 2$. Each step up the hierarchy multiplies the count by $\text{istep}^2 = 4$. The Instantiation axis multiplier thus sets the exponent step of the hierarchy, even though it does not limit the hierarchy's range to that of the vertex ladder itself. Whether this exponent-step correspondence is a theorem or a structural echo is left to Conjecture 9.5.

The singular origin. The unique $\pi/4$ path uses $(N, I, D, C) = (1, 2, 3, 5)$: the full axis vocabulary in order. Section 5.4 discusses the Fibonacci structure of this combination; from the axis-geometry perspective, it is simply the complete set of thologram-derived tokens assembled in a single convergent run, with zero generative entropy. No other combination of S -values in any algorithm reaches $\pi/4$.

What is proved vs. what is argued. The following bindings are established in Paper 1: the vertex values $2^0, 2^1, 2^2$; the axis multipliers $5, 3, 2$; the step $\Delta = \text{istep}^2 = 4$; and the seed triple $(1, 3, 5)$ for the $\pi/4$ branch. The following bindings are argued here on the basis of the sweep data and structural correspondence, but not yet proved as theorems: that the even-exponent ladder $0, 2, 4, 6, 8$ is the unique consequence of axis geometry; that the intermediate counts 2^6 and 2^4 for $\ln 2$ and e are forced by the axis structure rather than by algorithm-specific combinatorics alone; and that no non-axis-derived seed set reproduces the same gapless hierarchy. Conjecture 9.5 (Section 9) states the last of these formally.

The power-of-two pattern is therefore not merely an empirical curiosity about five classical series. It is the **binary vertex geometry of the thologram expressing itself combinatorially** when the axis-derived seed vocabulary is swept across the five tholonic branches. The hierarchy is tholonic not only in its algorithms and roles, but in the powers of two that count its convergent paths.

4.6 5. Generative Entropy and the Singularity of $\pi/4$

4.7 Definition

Define the *generative entropy* of a constant κ (with respect to the sweep) as the number of bits required to specify which seed combination produces κ :

$$H(\kappa) = \log_2(\text{number of seed combinations producing } \kappa).$$

Constant	Paths	$H(\kappa)$ (bits)
$\pi/4$	1	$\log_2(1) = 0$
e	16	$\log_2(16) = 4$
$\ln 2$	64	$\log_2(64) = 6$
φ	256	$\log_2(256) = 8$
$\sqrt{2}$	256	$\log_2(256) = 8$

4.7.1 5.2 $\pi/4$ has zero generative entropy

There is exactly one seed combination that produces $\pi/4$: $(N, I, D, C) = (1, 2, 3, 5)$ applied to the Leibniz algorithm. No information is needed to select it, because no choice exists. The path is uniquely determined.

This is a striking inversion of how π is ordinarily characterized. π is typically regarded as one of the most *representationally complex* constants: transcendental, infinite, non-repeating, without closed-form algebraic expression. Yet within this seed-space framework it is the most *ordered*, the most constrained, the most “prime-like” in the sense of being irreducible to simpler combinatorial constituents.

4.8 Two distinct kinds of complexity

The sweep reveals a distinction between two fundamentally different senses of complexity:

Dimension	$\pi/4$	$\varphi, \sqrt{2}$
Representational complexity	High: transcendental, infinite, non-repeating	Low: roots of minimal quadratics
Generative entropy	Zero: one unique path	Maximum: all 256 paths converge

$\pi/4$ is hard to *describe* but uniquely *located* in the seed space. φ and $\sqrt{2}$ are easy to *describe* algebraically but require more bits to specify *which* path was taken to reach them.

4.9 Uniqueness of the Fibonacci seeds

The one combination that produces $\pi/4$ is $(1, 2, 3, 5)$, the first four terms of the Fibonacci sequence. This is the same sequence whose ratio converges to φ . The seeds that uniquely unlock $\pi/4$ are simultaneously the seeds that encode the golden ratio's generative structure. This suggests a deep connection at the level of the generative apparatus, not merely at the level of the constants themselves.

4.10 6. φ and $\sqrt{2}$ as Universal Attractors

4.11 Shared properties

φ and $\sqrt{2}$ are the only constants in the hierarchy that occupy the same level (2^8). Every other level has exactly one occupant. Their shared universality reflects several concurrent structural facts:

Algebraic minimality. Both are roots of the simplest possible quadratics: - φ satisfies $x^2 - x - 1 = 0$ - $\sqrt{2}$ satisfies $x^2 - 2 = 0$

These are the two minimal quadratic irrationals: one more step above the integers than any quadratic can be.

Metallic-mean adjacency. Both constants also sit on adjacent rungs of the *metallic-mean* ladder. The metallic mean M_n is the positive root of $x^2 - nx - 1 = 0$, equivalently $M_n = (n + \sqrt{n^2 + 4})/2$. For $n = 1$, $M_1 = \varphi$ (the golden mean). For $n = 2$, $M_2 = 1 + \sqrt{2}$ (the silver mean). Substituting $y = x - 1$ into the silver-mean equation $(y + 1)^2 - 2(y + 1) - 1 = 0$ yields $y^2 - 2 = 0$, so $y = \sqrt{2}$. Thus $\sqrt{2} = M_2 - 1$: not itself a metallic mean, but the value one unit below the silver mean on the same ladder. φ occupies M_1 directly; $\sqrt{2}$ occupies $M_2 - 1$. The sisterhood is therefore not only algorithmic universality but also adjacency in the metallic-mean family at the first two indices.

Self-referential fixed-point structure. Both algorithms are self-referential in a deep sense: - $\varphi = 1 + 1/\varphi$ (the continued fraction contains itself) - $\sqrt{2} = (\sqrt{2} + 2/\sqrt{2})/2$ (the Babylonian average reconstructs itself)

This structural self-reference is what makes the algorithms converge from any positive starting point.

Geometric ubiquity. $\sqrt{2}$ is the diagonal of the unit square; φ is the diagonal-to-side ratio of the regular pentagon. Both appear throughout natural proportion, tiling, and growth patterns. Their geometric generality parallels their algorithmic universality.

4.12 A natural polarity

The hierarchy exhibits a natural polarity between the universal attractors and the singular origin:

- φ and $\sqrt{2}$ are **universal receivers**: present everywhere, reachable from any starting point, requiring no special key.

- $\pi/4$ is the **singular source**: present in exactly one place, requiring the precise Fibonacci key $(1, 2, 3, 5)$ to unlock it.
- e and $\ln 2$ are **intermediaries**: each a factor of 4 more accessible than the last, forming the cascade connecting source to receiver.

4.13 7. The Complete Cascade: Restoring π at the 2^2 Slot

4.14 The divide-by-four cascade

The seed-space counts form a cascade that divides by 4 at each step:

$$256 \xrightarrow{\div 4} 64 \xrightarrow{\div 4} 16 \xrightarrow{\div 4} 4 \xrightarrow{\div 4} 1$$

The raw sweep records 1 at the final position (for $\pi/4$). The predicted intermediate position of 4 appears unoccupied.

4.14.1 7.2 The tholonic Class A algorithm outputs $\pi/4$, not π

This apparent gap is resolved by attending to what the tholonic algorithm actually computes. The Class A branch for $\pi/4$ is derived from the axis geometry of the thologram: the seeds $(N_0, D_0, C_0) = (1, 3, 5)$ come from the axis multipliers and the generative unit, and the advance step $\Delta = \text{istep}^2 = 4$ is derived from the Instantiation axis (Paper 1, Section 2.1). The algorithm was not constructed from the Leibniz series; the Leibniz series is what the algorithm reduces to mathematically, as a consequence of its structure (Paper 1, Proposition 5.5).

The raw output of this algorithm is $\pi/4$. Recovering π requires an explicit $\times 4$ scaling step that is external to the algorithm itself. When that scaling is applied, the one seed combination producing $\pi/4$ corresponds to four combinations producing π (one per element of the $\times 4$ factor). The $2^2 = 4$ slot is not empty; it is occupied by π .

This is not a rescaling convention borrowed from the Leibniz literature. It is an observation about the tholonic algorithm's native output: the algorithm converges to $\pi/4$ as a structural consequence of its Class A geometry, and π itself sits one scaling step above that native value.

4.15 The complete hierarchy

Constant	Combinations	Power of 2
φ	256	2^8
$\sqrt{2}$	256	2^8
$\ln 2$	64	2^6
e	16	2^4
π	4	2^2

Constant	Combinations	Power of 2
$\pi/4$	1	2^0

Every even power of 2 from 2^0 to 2^8 is occupied. The sequence is complete and gapless:

$$\begin{array}{cccccc} 2^0 & 2^2 & 2^4 & 2^6 & 2^8 & 2^8 \\ \pi/4 & \pi & e & \ln 2 & \sqrt{2} & \varphi \end{array}$$

4.16 The four-stage emanation structure

Grouping by accessibility:

Stage	Constants	Combinations	Multiplier from previous stage
0 — Singular	$\pi/4$	1	(origin)
1 — Rare	e	16	$\times 16$
2 — Common	$\ln 2$	64	$\times 4$
3 — Universal	$\varphi, \sqrt{2}$	256	$\times 4$

Each outward step multiplies accessibility by 4. π occupies the slot between Stage 0 ($\pi/4$) and Stage 1 (e), at $\times 4$ from the origin.

4.17 8. π as Archetype Rather Than Instance

4.18 The apparent paradox

φ and $\sqrt{2}$ are the *most common* constants in this framework, appearing in every possible seed combination. Yet π is fundamental to far more of mathematics and physics than either of them: it appears in the normal distribution, the Fourier transform, the Basel problem ($1 + 1/4 + 1/9 + \dots = \pi^2/6$), wave equations, and the geometry of all circles. How can the rarest value in the seed space be the most fundamental value in mathematics?

4.19 The archetype-instance distinction

The resolution lies in the distinction between an *archetype* and an *instance*:

- An **instance** is common and reproducible. It appears *in* particular things. It is reachable from many starting points.
- An **archetype** is singular and irreducible. It is the pattern that generates instances rather than being one itself. It is reached only by a specific path.

By this distinction the hierarchy maps precisely:

Constant	Role
$\varphi, \sqrt{2}$	Instances: universal attractors, appear everywhere
$\ln 2, e$	Mediators: bridge between archetype and instance
π	Archetype: singular, irreducible, the source

4.19.1 8.3 Consistency with π 's mathematical behavior

This interpretation is consistent with how π actually behaves across mathematics. φ and $\sqrt{2}$ appear in *specific structures*: this spiral, this rectangle, this triangle. π appears in the *laws governing all structures*: in the relationship between circumference and diameter, in the structure of the normal distribution, in the completeness of the Fourier basis, in quantum mechanics. It shows up not in particular instances but in the framework within which all instances are measured.

An archetype does not need to be common. Its rarity is its power. If π appeared in all 256 combinations it would be just another attractor. The fact that it requires the one specific Fibonacci key (1, 2, 3, 5) suggests it is not arrived at by exploration or accident: every other constant can be stumbled upon from any starting point, but π must be sought.

5 Open Questions

The following conjectures are stated informally; resolving them would elevate the empirical observations of this paper to theorems.

Conjecture 9.1 (Power-of-two characterization). The seed set $\{1, 2, 3, 5\}$ is not the only set for which the frequency table of convergences is composed entirely of exact powers of two. Characterize all finite seed sets $S \subset \mathbb{R}_{>0}$ with this property.

Conjecture 9.2 (Gapless even-power sequences). For the five algorithms as defined, any seed set satisfying Conjecture 9.1 also produces a frequency table in which the exponents form a gapless sequence of even integers from 0 to $2 \log_2 |S^4|$. Prove or disprove.

Conjecture 9.3 (Fibonacci necessity for $\pi/4$). Among all seed sets satisfying Conjecture 9.1, the unique combination producing $\pi/4$ always consists of consecutive Fibonacci numbers. Prove or find a counterexample.

Conjecture 9.4 (Unlabeled convergences). The 687 runs converging to unlabeled values in Section 3.2 also exhibit a power-of-two frequency table when grouped by distinct convergent value. Verify computationally and characterize the unlabeled constants.

Conjecture 9.5 (Axis-geometry necessity). A finite seed set S produces the gapless even-exponent hierarchy $2^0, 2^2, \dots, 2^{2 \log_2 |S|}$ under the five Paper 1 algorithms if and only if

S is the axis-derived vocabulary $\{2^0, \text{istep}, m_C, m_D\} = \{1, 2, 3, 5\}$, where m_C and m_D are the Contribution and Definition axis multipliers. Prove or find a counterexample.

6 Conclusion

A computational sweep of all $4^4 = 256$ seed combinations from $\{1, 2, 3, 5\}$ across five tholonic algorithms reveals that the number of combinations converging to each classical constant is an exact power of two. The powers are $2^8, 2^8, 2^6, 2^4$, and 2^0 , and when the Leibniz algorithm's raw output $\pi/4$ is scaled by 4 to restore π , the slot $2^2 = 4$ is filled and the sequence of even powers from 2^0 to 2^8 is complete and gapless.

This cascade divides by 4 at each step and admits a clean entropy interpretation: $\pi/4$ has zero generative entropy (one path, no choice), while φ and $\sqrt{2}$ have maximum generative entropy (all paths lead there). The intermediate constants e and $\ln 2$ form a bridge, each a factor of 4 more accessible than the last.

The hierarchy mirrors the algebraic standing of the constants: quadratic irrationals at maximum accessibility, transcendentals at minimum accessibility. Section 4.5 further argues that the power-of-two structure is bound to the tholonic axis geometry: the sweep seed set $\{1, 2, 3, 5\}$ is axis-derived, the cascade rate $\div 4$ equals istep^2 , and the even-exponent ladder propagates the binary vertex structure through the Instantiation axis. We argue this is not coincidence but reflects what the constants structurally are within the tholonic framework, and that π 's position as the singular source in the seed space is consistent with its role as an archetype rather than an instance in the broader structure of mathematics.

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